The first smooth-absolute IRMF component

Sampling error, target shift, and biased signal error

4 Theoretical guarantees

The first IRMF component fits a local constant to noisy observations using Gaussian-smoothed absolute loss. We first bound the sampling error around the population target. We then bound the deterministic shift from that target to the biased signal, the output of the same filter on noiseless data. Adding these two terms gives the error relative to the biased signal.

The three fits and what their differences measure

Let $Y_u = X_u + \varepsilon_u$, where X_u is the clean signal and ε_u is observation noise. At a location t, use fixed weights $w_{u,t} \geq 0$ with positive total weight. All three fits use these same weights and the same loss

$$\rho_H(r) = \mathbf{E}_Z|r + HZ|, \qquad H > 0, \qquad Z \sim N(0,1),$$

where the auxiliary variable Z is independent of the observation noise. Define

$$\begin{aligned}
\widetilde{S}_t &:= \operatorname*{arg\,inf}_{z\in\mathbf{R}} \sum_u w_{u,t}\rho_H(Y_u - z), && \text{observed component,}\\
S_t &:= \operatorname*{arg\,inf}_{z\in\mathbf{R}} \sum_u w_{u,t}\mathbf{E}_\varepsilon\rho_H(X_u + \varepsilon_u - z), && \text{population target,}\\
X_t^{\mathrm{b}} &:= \operatorname*{arg\,inf}_{z\in\mathbf{R}} \sum_u w_{u,t}\rho_H(X_u - z), && \text{biased signal.}
\end{aligned}$$

The observed component $\widetilde{S}_t$ is what the algorithm computes from the noisy data. The biased signal X_t^{b} is what the same algorithm would compute if the clean signal were available. The population target S_t minimizes the noisy loss after that loss has been averaged over possible noise realizations.

Only $\widetilde{S}_t$ changes with the realized noise. For a fixed signal, design, and noise law, S_t and X_t^{b} are deterministic. Averaging a loss and then minimizing it is generally different from averaging its minimizer, so S_t need not equal $\mathbf{E}\widetilde{S}_t$.

$\widetilde{S}_t - S_t$**: sampling error.** The displacement of the observed fit from its population target. This finite-sample error need not have mean zero.

$S_t - X_t^{\mathrm{b}}$**: target shift.** The change in the minimizer caused by averaging over observation noise. It can be nonzero even for mean-zero noise because the loss is nonlinear.

$\widetilde{S}_t - X_t^{\mathrm{b}}$**: biased signal error.** The total change in the extracted component caused by adding observation noise to the clean signal.

These differences satisfy the exact identity

$$\boxed{\widetilde{S}_t - X_t^{\mathrm{b}} = \underbrace{\widetilde{S}_t - S_t}_{\text{sampling error}} + \underbrace{S_t - X_t^{\mathrm{b}}}_{\text{target shift}}.}$$

We call X_t^{b} the *biased signal* because local filtering can change the signal value even without observation noise. The difference $X_t^{\mathrm{b}} - X_t$ is deterministic filtering distortion; the name does not identify X_t^{b} with the expectation of an estimator. The biased signal error measures deviation from X_t^{b}, not from X_t. Recovering X_t would also require controlling $X_t^{\mathrm{b}} - X_t$.

From sampling error to biased signal error

Put $\Delta_t = S_t - X_t^{\mathrm{b}}$. The argument has three steps:

1. Bound the sampling error using the expansion

$$\widetilde{S}_t - S_t = F_t^{-1}\nabla_t + R_t, \qquad |\widetilde{S}_t - S_t| \le \mathcal{E}_x.$$

   Here F_t is the population curvature at S_t, ∇_t is the centered empirical score there, and R_t is the remainder. Proposition 4.1 gives the general expansion; Corollary 4.3 specifies $\mathcal{E}_x$ under Gaussian noise.

2. Bound the target shift, which contains no sampling fluctuation:

$$|\Delta_t| = |S_t - X_t^{\mathrm{b}}| \le \beta_t.$$

   Proposition 4.5 obtains this deterministic bound by comparing the score equations for the two deterministic targets.

3. Add the shift to obtain the biased signal error:

$$\widetilde{S}_t - X_t^{\mathrm{b}} = F_t^{-1}\nabla_t + R_t + \Delta_t, \qquad |\widetilde{S}_t - X_t^{\mathrm{b}}| \le \mathcal{E}_x + \beta_t.$$

   Corollary 4.6 combines the first two steps on the same probability event. Subtracting the oracle term $F_t^{-1}\nabla_t$ leaves $R_t + \Delta_t$, so it does not remove the target shift.

For orientation, let n be the number of observations, m the number of evaluation points, and h the spatial window radius. Set $z_x = x + \log(4m)$, where $x \ge 0$ sets the failure probability e^{-x}. For independent Gaussian noise, fixed loss and noise scales, regular windows, and normalized population curvature bounded away from zero,

$$\mathcal{E}_x = O\Big(\sqrt{z_x/(nh)}\Big), \qquad \mathcal{R}_x = O(z_x/(nh)),$$

where $\mathcal{R}_x$ bounds $\max_t |R_t|$. Corollary 4.4 gives sufficient local sample-size conditions. The shift Δ_t can persist as the sampling error decreases. All proofs are in Appendix A.

4.1 Sampling error: model and expansion

Let u range over n deterministic observation locations in $[0,1]$. Let $\mathcal{T}_n \subset [0,1]$ be a nonempty finite set of evaluation points and put $m = |\mathcal{T}_n|$. The signal, design, bandwidth, and weights are deterministic. For $t \in \mathcal{T}_n$, write

$$w_{u,t} := K_h(t-u), \qquad W_t := \sum_u w_{u,t}, \qquad Q_t^2 := \sum_u w_{u,t}^2, \qquad k_t := \max_u w_{u,t}. \tag{1}$$

Assume $w_{u,t} \geq 0$ and $W_t > 0$. Suppose that $Y_u = X_u + \varepsilon_u$, where the errors are independent and identically distributed with $\mathbf{E}|\varepsilon_u| < \infty$, and let

$$\phi(z) := \mathbf{E}\rho(z+\varepsilon). \tag{2}$$

The argument below also allows independent, non-identically distributed errors with finite first moments if $\phi(X_u - x)$ in each summand is replaced by $\mathbf{E}\rho(Y_u - x)$.

Assume that $\rho \in C^3(\mathbf{R})$ is convex and coercive, meaning $\rho(z) \to \infty$ as $|z| \to \infty$, and, for finite constants L_1, L_2, L_3,

$$|\rho'(z)| \leq L_1, \qquad 0 \leq \rho''(z) \leq L_2, \qquad |\rho'''(z)| \leq L_3, \qquad z \in \mathbf{R}. \tag{3}$$

For the Gaussian-smoothed absolute-value contrast, with a deterministic $H > 0$,

$$\begin{aligned} &\rho_H(r) := \mathbf{E}_Z|r + HZ|, \qquad Z \sim N(0,1), \\ &\rho_H'(r) = \operatorname{erf}\left(\frac{r}{\sqrt{2}H}\right), \qquad \rho_H''(r) = \frac{\sqrt{2/\pi}}{H} e^{-r^2/(2H^2)}, \\ &\rho_H'''(r) = -\frac{\sqrt{2/\pi}\, r}{H^3} e^{-r^2/(2H^2)}. \end{aligned} \tag{4}$$

One may take the conservative constants

$$L_1 = 1, \qquad L_2 = \frac{2}{H\sqrt{2\pi}}, \qquad L_3 = \frac{2}{H^2\sqrt{2\pi}}. \tag{5}$$

The Lipschitz bound $|\rho'| \leq L_1$ and the finite first noise moment make the expected objectives finite. The inequality $\mathbf{E}\rho(Y_u - z) \geq \rho(-z) - L_1\mathbf{E}|Y_u|$ also makes each population objective coercive. Thus both empirical and population minimizers exist. Define the empirical fit $\widetilde{S}_t$, its population target S_t, the population curvature F_t, and the centered score ∇_t by

$$\widetilde{S}_t := \operatorname*{arg\,inf}_{z \in \mathbf{R}} \sum_u \rho(Y_u - z) w_{u,t}, \qquad S_t := \operatorname*{arg\,inf}_{z \in \mathbf{R}} \sum_u \mathbf{E}\rho(Y_u - z) w_{u,t}, \tag{6}$$

$$F_t := \sum_u \phi''(X_u - S_t) w_{u,t}, \qquad D_t := F_t^{1/2}, \tag{7}$$

$$\nabla_t := \sum_u \left\{\rho'(Y_u - S_t) - \mathbf{E}\rho'(Y_u - S_t)\right\} w_{u,t}. \tag{8}$$

Also define the biased signal and target shift by

$$X_t^{\mathrm{b}} := \operatorname*{arg\,inf}_{z \in \mathbf{R}} \sum_u \rho(X_u - z) w_{u,t}, \qquad \Delta_t := S_t - X_t^{\mathrm{b}}. \tag{9}$$

Assume this minimizer is unique; this is automatic for ρ_H. The two signed errors satisfy the exact identity

$$\widetilde{S}_t - X_t^{\mathrm{b}} = (\widetilde{S}_t - S_t) + \Delta_t. \tag{10}$$

Require $F_t > 0$ for every $t \in \mathcal{T}_n$; convexity then makes the population minimizer unique. For ρ_H, all three objectives are strictly convex and $F_t > 0$ whenever $W_t > 0$. They need not be globally strongly convex: their curvatures tend to zero in the tails. The proof uses positive curvature near S_t, not a uniform curvature lower bound on $\mathbf{R}$.

The target S_t minimizes the expected noisy loss. It generally differs from X_t^{b}. No assumption that the first component equals the full signal value X_t is needed.

The constants below separate three controls: C_1 bounds the centered score, C_2 bounds the curvature fluctuation, and C_3 bounds the third derivative. The scale $D_t = \sqrt{F_t}$ expresses the first two bounds in local curvature units. For $x \geq 0$ and $z_x := x + \log(4m)$, set

$$C_1 := \max_{t \in \mathcal{T}_n} \frac{L_1}{D_t} \left\{ Q_t \sqrt{2z_x} + \frac{2k_t z_x}{3} \right\}, \tag{11}$$

$$C_2 := \max_{t \in \mathcal{T}_n} \frac{L_2}{D_t} \left\{ Q_t \sqrt{2z_x} + \frac{k_t z_x}{3} \right\}, \tag{12}$$

$$C_3 := \max_{t \in \mathcal{T}_n} \frac{L_3 W_t}{D_t^2}, \qquad D_{\min} := \min_{t \in \mathcal{T}_n} D_t, \qquad \delta := \frac{C_2}{D_{\min}}. \tag{13}$$

The parameter δ bounds the relative curvature error. The two conditions below keep the empirical curvature positive near S_t and make the scalar Taylor expansion valid at the empirical minimizer.

Proposition 4.1 (Sampling error expansion)**.** *Suppose that the conditions above hold and that*

$$\delta < 1, \qquad \frac{C_1 C_3}{D_{\min}(1-\delta)^2} < \frac{4}{9}. \tag{14}$$

Then there is a random set $\Omega(x)$ with $\mathbf{P}\{\Omega(x)\} \geq 1 - e^{-x}$ such that

$$\sup_{t \in \mathcal{T}_n} \left| D_t\big(\widetilde{S}_t - S_t - F_t^{-1}\nabla_t\big) \right| \leq \frac{1}{D_{\min}} \left\{ \frac{9C_1^2 C_3}{8(1-\delta)^3} + \frac{C_1 C_2}{1-\delta} \right\}. \tag{15}$$

On the same event, every empirical minimizer is unique and, for every t,

$$|\widetilde{S}_t - S_t| \leq \frac{3C_1}{2D_t(1-\delta)}. \tag{16}$$

Define $R_t = \widetilde{S}_t - S_t - F_t^{-1}\nabla_t$. The oracle-corrected estimator

$$\widetilde{S}_t^{\mathrm{corr}} := \widetilde{S}_t - F_t^{-1}\nabla_t \tag{17}$$

satisfies $\widetilde{S}_t^{\mathrm{corr}} - S_t = R_t$, with the remainder bound (15)*.*

Remark (Sign of the correction)**.** *With ∇_t defined by* (8)*, the minus sign in* (15) *is forced by differentiation: $f_\circ'(S_t) = -\nabla_t$. Thus $\widetilde{S}_t - S_t$ has first-order term $F_t^{-1}\nabla_t$, and one subtracts that term to correct $\widetilde{S}_t$. This removes the leading sampling fluctuation; it does not remove Δ_t.*

4.2 Score and curvature bounds

The next lemma supplies the probabilistic input to Proposition 4.1. Its score and curvature bounds use Bernstein's inequality and a union bound over the finite evaluation grid. The third derivative bounds are deterministic.

Fix $t \in \mathcal{T}_n$. All quantities in this subsection may depend on t. For the proof, abbreviate the population and empirical objectives as

$$f(z) := \sum_u \phi(X_u - z) w_{u,t}, \qquad f_\circ(z) := \sum_u \rho(Y_u - z) w_{u,t}, \tag{18}$$

and

$$x^* := \arg\inf_z f(z) = S_t, \qquad x_\circ := \arg\inf_z f_\circ(z) = \widetilde{S}_t. \tag{19}$$

Further, put

$$F_\circ := f_\circ''(x^*), \qquad F := f''(x^*) = D^2. \tag{20}$$

For $r > 0$, let $U_t(r) := \{z \in \mathbf{R} : |D(z - x^*)| \le r\}$.

Lemma 4.2. *There is an event $\Omega(x)$ of probability at least $1 - e^{-x}$ on which, simultaneously for every $t \in \mathcal{T}_n$,*

(i)

$$\frac{1}{D}|f'(x^*) - f_\circ'(x^*)| = \frac{1}{D}\left|\sum_u \left\{\rho'(Y_u - x^*) - \mathbf{E}\rho'(Y_u - x^*)\right\} w_{u,t}\right| \le C_1, \tag{21}$$

$$\frac{1}{D}|f''(x^*) - f_\circ''(x^*)| = \frac{1}{D}\left|\sum_u \left\{\rho''(Y_u - x^*) - \mathbf{E}\rho''(Y_u - x^*)\right\} w_{u,t}\right| \le C_2. \tag{22}$$

(ii)

$$\sup_{z \in \mathbf{R}} \frac{1}{D^2}\left|\sum_u \rho'''(Y_u - z) w_{u,t}\right| \le C_3. \tag{23}$$

(iii) For every $r > 0$,

$$\sup_{z \in U_t(r)} \frac{1}{D^2}\left|\sum_u \left\{\phi''(X_u - z) - \phi''(X_u - x^*)\right\} w_{u,t}\right| \le \frac{C_3 r}{D}. \tag{24}$$

The proof in Appendix A.2 uses only the bounds in Lemma 4.2. It also applies with other constants satisfying those bounds on a specified event. Gaussian noise permits sharper variance bounds.

4.3 Gaussian noise and normalized weights

Multiplying all weights in a window by a positive constant changes neither minimizer nor $F_t^{-1}\nabla_t$. Define

$$a_{u,t} := \frac{w_{u,t}}{W_t}, \qquad N := \min_t \frac{1}{\sum_u a_{u,t}^2}, \qquad \alpha := \max_{t,u} a_{u,t}. \tag{25}$$

Here N is the smallest effective sample size across windows. The population curvature and score for the normalized objective are $\bar{F}_t = F_t/W_t$ and $\bar{\nabla}_t = \nabla_t/W_t$. Their ratio is still $F_t^{-1}\nabla_t$.

Assume now $\varepsilon_u \sim N(0, \sigma^2)$ independently, with $\sigma \ge 0$, and $\rho = \rho_H$. Put

$$c := \sqrt{2/\pi}, \qquad B := \sqrt{H^2 + \sigma^2}. \tag{26}$$

Independence of the Gaussian variables in the two expectations gives the exact identity

$$\phi(r) = \mathbf{E}_\varepsilon \rho_H(r+\varepsilon) = \rho_B(r). \tag{27}$$

Thus X_t^{b} fits the clean signal at scale H, whereas S_t fits it at scale B. In particular,

$$\sum_u a_{u,t} \operatorname{erf}\left(\frac{X_u - X_t^{\mathrm{b}}}{\sqrt{2}H}\right) = 0,$$

and the population target and normalized curvature satisfy

$$\sum_u a_{u,t} \operatorname{erf}\left(\frac{X_u - S_t}{\sqrt{2}B}\right) = 0, \qquad \bar{F}_t = \frac{c}{B}\sum_u a_{u,t} e^{-(X_u - S_t)^2/(2B^2)}. \tag{28}$$

The weighted score is positive below all active signal values and negative above them. Its unique zero S_t therefore lies between the smallest and largest X_u having positive weight. If

$$A \geq \max_t \max_{u,v:\, a_{u,t}a_{v,t}>0} |X_u - X_v|, \qquad \lambda := \frac{c}{B} e^{-A^2/(2B^2)}, \tag{29}$$

then $\bar{F}_t \geq \lambda > 0$ at every evaluation point. More generally, any known positive lower bound $\lambda \leq \min_t \bar{F}_t$ can be used below.

The bounds s_1 and s_2 below control the normalized score and curvature fluctuations. Dividing s_2 by the curvature lower bound λ gives the relative perturbation d. For $L_2 = c/H$ and $L_3 = c/H^2$, define

$$\begin{aligned} v_1 &:= \min\{1, \sigma L_2\}, & v_2 &:= \min\{L_2, \sigma L_3\}, \\ s_1 &:= v_1\sqrt{\frac{2z_x}{N}} + \frac{2\alpha z_x}{3}, & s_2 &:= v_2\sqrt{\frac{2z_x}{N}} + \frac{L_2\alpha z_x}{3}, \\ d &:= \frac{s_2}{\lambda}, & z_x &= x + \log(4m). \end{aligned} \tag{30}$$

Corollary 4.3 (Explicit Gaussian bounds)**.** *Under the Gaussian model and the deterministic design conditions above, suppose*

$$d < 1, \qquad \frac{s_1 L_3}{\lambda^2(1-d)^2} < \frac{4}{9}. \tag{31}$$

With probability at least $1 - e^{-x}$, simultaneously for all t,

$$|\widetilde{S}_t - S_t| \leq \mathcal{E}_x := \frac{3s_1}{2\lambda(1-d)}, \tag{32}$$

$$|F_t^{-1}\nabla_t| \leq \frac{s_1}{\lambda}, \tag{33}$$

$$|R_t| \leq \mathcal{R}_x := \frac{9L_3 s_1^2}{8\lambda^3(1-d)^3} + \frac{s_1 s_2}{\lambda^2(1-d)}, \tag{34}$$

$$R_t := \widetilde{S}_t - S_t - F_t^{-1}\nabla_t.$$

In particular,

$$|\widetilde{S}_t^{\mathrm{corr}} - S_t| \leq \mathcal{R}_x. \tag{35}$$

The constants $\mathcal{E}_x$ and $\mathcal{R}_x$ are specified by the weights, x, H, σ, and a curvature lower bound such as (29)*. These bounds concern the sampling error. The target shift will enter only when we change the reference to X_t^{b} in Section 4.6.*

Remark (Dependence on the noise and loss scales)**.** *The noise scale enters through* $B = \sqrt{H^2 + \sigma^2}$*, the lower bound* λ*, and the variance bounds* v_1, v_2*. These are sufficient, conservative constants. At* $\sigma = 0$*, the empirical and population objectives coincide with the clean objective, so both errors and* R_t *vanish; the Bernstein linear terms need not vanish. If* H *varies with sample size, the explicit conditions must be checked again:* L_2 *and* L_3 *grow as* H^{-1} *and* H^{-2} *when* H *decreases.*

4.4 Bandwidth and sample-size conditions

Suppose that, for a known constant $C_K > 0$,

$$N \geq \frac{nh}{C_K}, \qquad \alpha \leq \frac{C_K}{nh}. \tag{36}$$

These assumptions say that the effective sample size is at least a constant times nh, and no individual observation has disproportionate weight. They can be checked directly from the normalized weights.

For an equally spaced design, the calculation is clearest in terms of the actual grid spacing Δ. Use untruncated interior windows, or a periodic grid with each observation counted once, and take $0 < h < 1/2$. Let $w_{u,t} = K((t-u)/h)$, using local offsets in the periodic case, with K supported on $[-1, 1]$, $0 \leq K \leq K_{\max}$, and $K(v) \geq k_0 > 0$ on $|v| \leq \eta$, where $0 < \eta < 1$. If $\eta h \geq \Delta$, the central interval has length $2\eta h$ and contains at least

$$\left\lfloor \frac{2\eta h}{\Delta} \right\rfloor \geq \frac{\eta h}{\Delta}$$

grid points. Consequently,

$$W_t \geq \frac{k_0 \eta h}{\Delta}, \qquad \max_u a_{u,t} \leq \frac{K_{\max}\Delta}{k_0 \eta h}, \qquad \sum_u a_{u,t}^2 \leq \max_u a_{u,t}.$$

For the usual unit-interval grids, $\Delta = 1/n$ when the right endpoint is excluded, and $\Delta = 1/(n-1)$ when both endpoints are included. For $n \geq 2$, both satisfy $\Delta \leq 2/n$. Thus the single choice

$$C_K = \frac{2K_{\max}}{k_0 \eta}$$

establishes (36) for either convention, provided $\eta h \geq \Delta$. The condition $nh \geq 2/\eta$ is sufficient. If the spacing is exactly $1/n$, the sharper constant $C_K = K_{\max}/(k_0\eta)$ and condition $nh \geq 1/\eta$ suffice. A periodic grid has n distinct locations per period; it does not count both endpoints as separate locations. There are at most $2h/\Delta + 1$ active observations, which also bounds the effective sample size from above. Both grid conventions therefore give effective sample size of order nh.

Set

$$q_n := \sqrt{\frac{C_K z_x}{nh}}, \qquad A_1 := \sqrt{2} v_1 + \frac{2}{3}, \qquad A_2 := \sqrt{2} v_2 + \frac{L_2}{3}. \tag{37}$$

Corollary 4.4 (A sufficient local sample size)**.** *Under the Gaussian model, the curvature lower bound, and* (36)*, the following condition implies* (31)*:*

$$nh > C_K z_x \max\left\{1, \frac{4A_2^2}{\lambda^2}, \frac{81 A_1^2 L_3^2}{\lambda^4}\right\}. \tag{38}$$

With probability at least $1 - e^{-x}$,

$$\max_t |\widetilde{S}_t - S_t| \le \frac{3A_1}{\lambda} q_n, \tag{39}$$

$$\max_t |R_t| \le \left\{ \frac{9L_3 A_1^2}{\lambda^3} + \frac{2A_1 A_2}{\lambda^2} \right\} q_n^2. \tag{40}$$

For fixed $H > 0$, fixed σ, bounded local signal oscillation, and fixed kernel constants, λ stays bounded away from zero. The two bounds then have orders $\sqrt{(x + \log(4m))/(nh)}$ and $(x + \log(4m))/(nh)$, respectively. The smallness conditions hold eventually if $(x + \log(4m))/(nh) \to 0$. These are upper bounds on sampling error; they do not assert equality of the remainder and the squared realized error. The biased signal error also contains Δ_t, which need not decrease when more observations are added at a fixed filtering scale.

4.5 Target shift: comparing the deterministic targets

The preceding bounds isolate the sampling fluctuation. A bound on $\widetilde{S}_t - X_t^{\mathrm{b}}$ also needs a bound on $S_t - X_t^{\mathrm{b}}$. Gaussian averaging changes the loss scale from H to B, so this shift can persist even when the sampling fluctuation becomes small.

Let the active signal range and its curvature bound be

$$A_t := \max_{u: a_{u,t} > 0} X_u - \min_{u: a_{u,t} > 0} X_u, \qquad \kappa_t := \frac{c}{B} e^{-A_t^2/(2B^2)}. \tag{41}$$

Both S_t and X_t^{b} lie in this signal range. Throughout the interval between them, the normalized population curvature is at least κ_t. This interval bound is stronger than knowing the curvature only at S_t.

Proposition 4.5 (Deterministic target shift)**.** *Under the Gaussian model, put*

$$\beta_t := \min\left\{ A_t, \frac{L_3 \sigma^2}{2\kappa_t} \right\}, \qquad \beta := \max_t \beta_t. \tag{42}$$

Then, deterministically,

$$|S_t - X_t^{\mathrm{b}}| \le \beta_t \qquad \text{for every } t. \tag{43}$$

One may take $\lambda = \min_t \kappa_t > 0$ *in Corollary 4.3.*

For fixed signal, weights, and H, the shift bound is $O(\sigma^2)$ as $\sigma \to 0$. It is a sufficient upper bound and can be conservative. At $\sigma = 0$, both errors vanish exactly. If the weighted signal values are symmetric about a number μ_t, both target scores vanish there, so $X_t^{\mathrm{b}} = S_t = \mu_t$ even when $\sigma > 0$. This includes constant signals. Symmetric observation locations alone do not imply symmetry of signal values for a curved signal.

Remark (An asymmetric window)**.** *Take equally weighted signal values* $(0, 0, 3)$, $H = 0.5$, *and* $\sigma = 1$. *The target at scale* b *is the unique root on* $[0, 3]$ *of*

$$2\,\mathrm{erf}\left(\frac{-z}{\sqrt{2}b}\right) + \mathrm{erf}\left(\frac{3 - z}{\sqrt{2}b}\right) = 0.$$

Bisection at $b = H$ *and* $b = B = \sqrt{1.25}$ *gives*

$$X_t^{\mathrm{b}} \approx 0.337245, \qquad S_t \approx 0.718197, \qquad \Delta_t \approx 0.380952.$$

Repeating each signal value r *times, with weights* $1/(3r)$ *and independent observation noise, preserves both targets and gives effective sample size* $3r$. *As* $r \to \infty$, *the sampling error tends to zero in probability, while the biased signal error tends to* Δ_t.

4.6 Biased signal error: adding the target shift

The sampling error and target shift now have separate bounds. The identity (10) combines them:

$$\boxed{\widetilde{S}_t - X_t^{\mathrm{b}} = \underbrace{F_t^{-1}\nabla_t + R_t}_{\text{sampling error}} + \underbrace{\Delta_t}_{\text{target shift}}}. \tag{44}$$

The same shift appears after oracle correction: $\widetilde{S}_t^{\mathrm{corr}} - X_t^{\mathrm{b}} = R_t + \Delta_t$.

Corollary 4.6 (Combined error bounds)**.** *Under the Gaussian model, if the smallness conditions of Corollary 4.3 hold, then with probability at least* $1 - e^{-x}$*, simultaneously for all* t*,*

$$|\widetilde{S}_t - S_t| \le \mathcal{E}_x, \tag{45}$$

$$|\widetilde{S}_t - X_t^{\mathrm{b}}| \le \beta_t + \mathcal{E}_x, \tag{46}$$

$$|\widetilde{S}_t^{\mathrm{corr}} - S_t| \le \mathcal{R}_x, \tag{47}$$

$$|\widetilde{S}_t^{\mathrm{corr}} - X_t^{\mathrm{b}}| \le \beta_t + \mathcal{R}_x. \tag{48}$$

Replacing β_t *by* β *gives uniform bounds over the evaluation grid. The shift bound is deterministic, so adding it does not change the failure probability.*

Under the additional conditions of Corollary 4.4, the same argument gives, with probability at least $1 - e^{-x}$,

$$\max_t |\widetilde{S}_t - X_t^{\mathrm{b}}| \le \beta + \frac{3A_1}{\lambda} q_n, \tag{49}$$

$$\max_t |\widetilde{S}_t^{\mathrm{corr}} - X_t^{\mathrm{b}}| \le \beta + \left\{ \frac{9L_3 A_1^2}{\lambda^3} + \frac{2A_1 A_2}{\lambda^2} \right\} q_n^2. \tag{50}$$

Thus the sampling contribution decreases at the stated rates, while the deterministic target shift remains.

For any realization, the reverse triangle inequality also gives

$$\left| \max_t |\widetilde{S}_t - X_t^{\mathrm{b}}| - \max_t |\Delta_t| \right| \le \max_t |\widetilde{S}_t - S_t|. \tag{51}$$

Thus a nonzero shift remains as sampling error decreases. The upper bound β itself is not a lower bound on the error.

4.7 Scope of the conclusions

All uniform probability statements concern the finite evaluation grid. A supremum over the continuum $[0, 1]$ requires additional regularity and control of the family of weighted processes. Adaptive choices of bandwidth or loss scale also require simultaneous control over those choices.

The oracle correction uses the unknown population target, noise law, and curvature. It is an expansion tool, not an estimator computable from the observations alone. Its remainder bound around S_t becomes a bound around X_t^{b} only after adding the target shift. Neither signed error is guaranteed to have the smaller absolute value on every realization.

In synthetic experiments, compute X^{b} by applying the filter to the clean signal at scale H, and compute S at scale B under Gaussian noise. Evaluate both errors on the same

noise realization and report $S - X^{\mathrm{b}}$. Comparing with S tests the stochastic expansion; comparing with X^{b} measures how noise changes the biased signal. Neither comparison alone establishes recovery of latent generative components.

The general proposition allows independent non-Gaussian errors with a finite first moment. The explicit Gaussian formulas for the population target and its shift do not cover arbitrary non-Gaussian noise. Later IRMF components use estimated residuals that share observations, so their dependent errors require a separate argument.

Throughout, F_t is evaluated at S_t. It cannot be replaced by a curvature evaluated at X_t^{b} without bounding their difference.

A Proofs

The argument has two stages. Lemma 4.2 controls the random score and curvature at the population target. Conditional on that event, the proof of Proposition 4.1 is deterministic: localize the empirical minimizer, apply scalar Taylor expansion, and replace empirical curvature by population curvature. The remaining proofs specialize these bounds and control the target shift.

A.1 Score and curvature concentration

Proof of Lemma 4.2. Differentiating (18) gives

$$f'(z) = -\sum_u \phi'(X_u - z)w_{u,t}, \qquad f'_\circ(z) = -\sum_u \rho'(Y_u - z)w_{u,t},$$
$$f''(z) = \sum_u \phi''(X_u - z)w_{u,t}, \qquad f''_\circ(z) = \sum_u \rho''(Y_u - z)w_{u,t}.$$

Since differentiation under the expectation is justified by (3), $\phi^{(j)}(a) = \mathbf{E}\rho^{(j)}(a + \varepsilon)$ for $j = 1, 2, 3$. The identities in part (i) follow.

For the first identity, set

$$Z_{u,t} := w_{u,t}\left\{\rho'(Y_u - x^*) - \mathbf{E}\rho'(Y_u - x^*)\right\}.$$

These variables are independent and centered. Moreover,

$$|Z_{u,t}| \le 2L_1 k_t, \qquad \sum_u \mathbf{E}Z_{u,t}^2 \le L_1^2 Q_t^2.$$

Bernstein's inequality therefore yields, for every $z > 0$,

$$\mathbf{P}\left(\left|\sum_u Z_{u,t}\right| > L_1\left\{Q_t\sqrt{2z} + \frac{2k_t z}{3}\right\}\right) \le 2e^{-z}. \tag{52}$$

For the second identity, use

$$\widetilde{Z}_{u,t} := w_{u,t}\left\{\rho''(Y_u - x^*) - \mathbf{E}\rho''(Y_u - x^*)\right\}.$$

Convexity and (3) give

$$|\widetilde{Z}_{u,t}| \le L_2 k_t, \qquad \sum_u \mathbf{E}\widetilde{Z}_{u,t}^2 \le L_2^2 Q_t^2.$$

Hence

$$\mathbf{P}\left(\left|\sum_u \widetilde{Z}_{u,t}\right| > L_2\left\{Q_t\sqrt{2z} + \frac{k_t z}{3}\right\}\right) \le 2e^{-z}. \tag{53}$$

Take $z = z_x = x + \log(4m)$ in (52) and (53), and take the union over the two inequalities and over $t \in \mathcal{T}_n$. The exceptional probability is at most $4me^{-z_x} = e^{-x}$. Equations (11) and (12) now prove part (i).

Part (ii) is deterministic. By (3) and (13),

$$\sup_{z\in\mathbf{R}} \frac{1}{D^2}\left|\sum_u \rho'''(Y_u - z)w_{u,t}\right| \le \frac{L_3 W_t}{D^2} \le C_3.$$

Finally, $|\phi'''| \le L_3$. The mean value theorem and $|z - x^*| \le r/D$ for $z \in U_t(r)$ give

$$\frac{1}{D^2}\left|\sum_u \left\{\phi''(X_u - z) - \phi''(X_u - x^*)\right\} w_{u,t}\right| \le \frac{L_3 W_t}{D^2}|z - x^*| \le \frac{C_3 r}{D},$$

which proves part (iii). □

A.2 Scalar localization and the remainder

Proof of Proposition 4.1. Fix $t \in \mathcal{T}_n$. Since $f'(x^*) = 0$,

$$f'_\circ(x^*) = -\sum_u \left\{\rho'(Y_u - x^*) - \mathbf{E}\rho'(Y_u - x^*)\right\} w_{u,t} = -\nabla_t. \tag{54}$$

Lemma 4.2(i) also gives

$$\frac{|F_\circ - F|}{F} \le \frac{C_2}{D} \le \delta. \tag{55}$$

In particular, $F_\circ \ge (1-\delta)F > 0$. With

$$b_\circ := \frac{D}{F_\circ}|f'_\circ(x^*)|, \tag{56}$$

the score bound and (55) imply

$$b_\circ \le \frac{C_1}{1-\delta}. \tag{57}$$

Use the scaled coordinate $v = D(z - x^*)$, and put

$$g(v) := f_\circ(x^* + v/D), \qquad a := g''(0) = F_\circ/D^2, \qquad \tau := C_3/D. \tag{58}$$

Then $a \ge 1 - \delta$, $|g'''(v)| \le \tau$, and $b_\circ = |g'(0)|/a$. Consequently,

$$\frac{\tau b_\circ}{a} \le \frac{C_1 C_3}{D(1-\delta)^2} < \frac{4}{9}. \tag{59}$$

If $b_\circ = 0$, convexity and $g''(0) > 0$ imply that 0 is the unique minimizer, and the claimed bounds are immediate. Otherwise set $r = 3b_\circ/2$. Taylor expansion of g' at zero yields

$$|g'(v) - g'(0) - av| \le \frac{\tau}{2}v^2.$$

Since $|g'(0)| = ab_\circ$,

$$g'(r) \geq -ab_\circ + ar - \frac{\tau r^2}{2} = \frac{ab_\circ}{2} - \frac{9\tau b_\circ^2}{8} > 0,$$
$$g'(-r) \leq ab_\circ - ar + \frac{\tau r^2}{2} = -\frac{ab_\circ}{2} + \frac{9\tau b_\circ^2}{8} < 0.$$

Moreover, for $|v| \leq r$,

$$g''(v) \geq a - \tau r > \frac{a}{3} > 0.$$

There is therefore a unique zero $v_\circ \in (-r, r)$ of g'. Convexity on $\mathbf{R}$ makes it the unique global minimizer: the derivative is negative to the left of this interval and positive to its right. This proves

$$|v_\circ| = D|x_\circ - x^*| \leq \frac{3}{2} b_\circ \leq \frac{3C_1}{2(1-\delta)}. \tag{60}$$

At the zero of g', the same Taylor expansion gives

$$\left| a\left(v_\circ + \frac{g'(0)}{a} \right) \right| \leq \frac{\tau}{2} v_\circ^2 \leq \frac{9\tau}{8} b_\circ^2. \tag{61}$$

Equivalently, the left-hand side is $D^{-1}F_\circ|x_\circ - x^* + F_\circ^{-1} f'_\circ(x^*)|$. Using (55) and (57), we obtain

$$\left| x_\circ - x^* + F_\circ^{-1} f'_\circ(x^*) \right| \leq \frac{9C_1^2 C_3}{8D^2(1-\delta)^3}. \tag{62}$$

It remains to replace $F_\circ$ by the population curvature F. By (55), (21), and $f'(x^*) = 0$,

$$\begin{aligned} \left| F_\circ^{-1} f'_\circ(x^*) - F^{-1} f'_\circ(x^*) \right| &= \frac{|F_\circ - F|}{F_\circ F} |f'_\circ(x^*)| \\ &\leq \frac{C_1 C_2}{D^2(1-\delta)}. \end{aligned} \tag{63}$$

The triangle inequality applied to (62) and (63), followed by (54), proves

$$D\left| x_\circ - x^* - F^{-1}\nabla_t \right| \leq \frac{1}{D}\left\{ \frac{9C_1^2 C_3}{8(1-\delta)^3} + \frac{C_1 C_2}{1-\delta} \right\}. \tag{64}$$

Taking the supremum over $t \in \mathcal{T}_n$ and using $D_t \geq D_{\min}$ gives (15). Subtracting $F_t^{-1}\nabla_t$ gives $\widetilde{S}_t^{\mathrm{corr}} - S_t = R_t$. □

Remark (The remainder coefficient)**.** *The coefficient* $9/8$ *is* $\frac{1}{2}(\frac{3}{2})^2$*. The factor* $1/2$ *comes from Taylor's theorem and* $3/2$ *from the localization radius.*

A.3 Gaussian constants

Proof of Corollary 4.3. The identity (27) follows because $\varepsilon + HZ \sim N(0, B^2)$; the target and curvature formulas follow by differentiation. For a Lipschitz function G and an independent copy ε' of ε,

$$\mathrm{Var}\{G(X_u + \varepsilon)\} = \tfrac{1}{2}\mathbf{E}\{G(X_u + \varepsilon) - G(X_u + \varepsilon')\}^2 \leq \mathrm{Lip}(G)^2 \sigma^2.$$

Apply this with $G(y) = \rho'_H(y - S_t)$ and $G(y) = \rho''_H(y - S_t)$. Their Lipschitz constants are at most L_2 and L_3. Combining this estimate with the boundedness estimates from

Lemma 4.2 gives variance bounds v_1^2 and v_2^2. Bernstein's inequality and the same union bound therefore give an event of probability at least $1 - e^{-x}$ on which

$$\max_t |\bar{\nabla}_t| \le s_1, \qquad \max_t |\bar{F}_{\circ,t} - \bar{F}_t| \le s_2, \qquad \bar{F}_{\circ,t} = \sum_u a_{u,t} \rho''_H(Y_u - S_t).$$

For the normalized objectives, the deterministic proof of Proposition 4.1 applies with the bounds

$$C_1 = \frac{s_1}{\sqrt{\lambda}}, \qquad C_2 = \frac{s_2}{\sqrt{\lambda}}, \qquad C_3 = \frac{L_3}{\lambda}, \qquad D_{\min} \ge \sqrt{\lambda}, \qquad \delta \le d.$$

Condition (31) is sufficient for (14). Substitution in (16), using $D_t \ge \sqrt{\lambda}$, proves the raw bound. Substitution in (64), followed by division by $D_t \ge \sqrt{\lambda}$, proves the remainder bound. The score bound and $\bar{F}_t \ge \lambda$ give (33). The identity $\widetilde{S}_t^{\mathrm{corr}} - S_t = R_t$ gives (35). □

A.4 Sufficient local sample size

Proof of Corollary 4.4. The mass bounds imply

$$s_1 \le \sqrt{2} v_1 q_n + \tfrac{2}{3} q_n^2, \qquad s_2 \le \sqrt{2} v_2 q_n + \tfrac{L_2}{3} q_n^2.$$

Condition (38) implies

$$q_n < \min\left\{1, \frac{\lambda}{2A_2}, \frac{\lambda^2}{9A_1 L_3}\right\}.$$

Therefore $s_1 \le A_1 q_n$, $s_2 \le A_2 q_n$, $d < 1/2$, and

$$\frac{s_1 L_3}{\lambda^2 (1-d)^2} \le \frac{4 A_1 q_n L_3}{\lambda^2} < \frac{4}{9}.$$

Apply Corollary 4.3 and use $(1-d)^{-1} \le 2$ and $(1-d)^{-3} \le 8$. □

A.5 Target shift and biased signal error

Proof of Proposition 4.5. Write

$$\psi_b(r) = \operatorname{erf}\left(\frac{r}{\sqrt{2}b}\right), \qquad G_{b,t}(z) = \sum_u a_{u,t} \psi_b(X_u - z).$$

Both scores are strictly decreasing, positive below the active signal range, and negative above it. Their zeros X_t^{b} and S_t therefore lie in that range. In particular, $|S_t - X_t^{\mathrm{b}}| \le A_t$.

Differentiating the Gaussian convolution identity gives $\psi_B(r) = \mathbf{E}\psi_H(r + \varepsilon)$. Since $\sup_r |\psi''_H(r)| = \sup_r |\rho'''_H(r)| \le L_3$, Taylor's theorem gives

$$|\psi_H(r+e) - \psi_H(r) - e\psi'_H(r)| \le \frac{L_3}{2} e^2 \qquad (r, e \in \mathbf{R}).$$

Taking expectation with $e = \varepsilon$, and using $\mathbf{E}\varepsilon = 0$ and $\mathbf{E}\varepsilon^2 = \sigma^2$, yields

$$|\psi_B(r) - \psi_H(r)| \le \frac{L_3 \sigma^2}{2}.$$

As $G_{H,t}(X_t^{\mathrm{b}}) = 0$ and the weights sum to one,

$$|G_{B,t}(X_t^{\mathrm{b}})| = \left|\sum_u a_{u,t}\{\psi_B(X_u - X_t^{\mathrm{b}}) - \psi_H(X_u - X_t^{\mathrm{b}})\}\right| \le \frac{L_3 \sigma^2}{2}. \tag{65}$$

For every z between X_t^{b} and S_t, active residuals satisfy $|X_u - z| \le A_t$. Hence

$$-G'_{B,t}(z) = \frac{c}{B} \sum_u a_{u,t} e^{-(X_u - z)^2/(2B^2)} \ge \kappa_t. \tag{66}$$

Since $G_{B,t}(S_t) = 0$, the mean value theorem gives

$$|S_t - X_t^{\mathrm{b}}| \le \frac{|G_{B,t}(X_t^{\mathrm{b}})|}{\kappa_t} \le \frac{L_3 \sigma^2}{2\kappa_t}.$$

Combining this with the range bound proves (43). The same curvature bound at S_t gives $\bar{F}_t \ge \kappa_t$, so $\lambda = \min_t \kappa_t$ is a valid choice in Corollary 4.3.

□

Proof of Corollary 4.6 and the rate bounds. On the event of Corollary 4.3, add the deterministic bound $|\Delta_t| \le \beta_t$ to (32) and to the corrected remainder bound. Identity (44) and the triangle inequality give (45)–(48) without changing the exceptional probability. Adding β to the bounds in Corollary 4.4 gives (49) and (50). □